Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using content-based filtering approach	
Experiment No: 11	**Date:**	**Enrolment No:92301733057**

Aim: To recommend an item to the user using content-based filtering approach

IDE: Google Colab

Theory:

There are a lot of applications where websites collect data from their users and use that data to predict the likes and dislikes of their users. This allows them to recommend the content that they like. Recommender systems are a way of suggesting or similar items and ideas to a user's specific way of thinking.

Recommender System is different types:

- **Collaborative Filtering:** Collaborative Filtering recommends items based on similarity measures between users and/or items. The basic assumption behind the algorithm is that users with similar interests have common preferences.
- **Content-Based Recommendation:** It is supervised machine learning used to induce a classifier to discriminate between interesting and uninteresting items for the user.

Content-Based Recommendation System: Content-Based systems recommends items to the customer similar to previously high-rated items by the customer. It uses the features and properties of the item. From these properties, it can calculate the similarity between the items.
In a content-based recommendation system, first, we need to create a profile for each item, which represents the properties of those items. From the user profiles are inferred for a particular user. We use these user profiles to recommend the items to the users from the catalog.

Item profile:
In a content-based recommendation system, we need to build a profile for each item, which contains the important properties of each item. For Example, If the movie is an item, then its actors, director, release year, and genre are its important properties, and for the document, the important property is the type of content and set of important words in it.
Let's have a look at how to create an item profile. First, we need to perform the TF-IDF vectorizer, here TF (term frequency) of a word is the number of times it appears in a document and The IDF (inverse document frequency) of a word is the measure of how significant that term is in the whole corpus.

User profile:
The user profile is a vector that describes the user preference. During the creation of the user's profile, we use a utility matrix that describes the relationship between user and item. From this information, the best estimate we can decide which item the user likes, is some aggregation of the profiles of those items.
Advantages and Disadvantages:

- **Advantages:**
    - No need for data on other users when applying to similar users.
    - Able to recommend to users with unique tastes.
    - Able to recommend new & popular items

- Explanations for recommended items.

- **Disadvantages**:
  - Finding the appropriate feature is hard.
  - Doesn't recommend items outside the user profile.

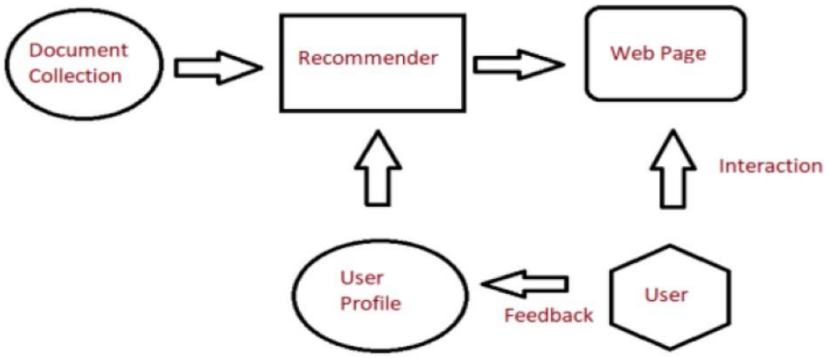

Fig: Recommender System

Pre Lab Exercise:

1. When can you use content-based recommendation system approach?

Content-based recommendation systems are used when recommendations need to be made based on a user's past preferences and the features of items. This approach is suitable when we have detailed information about items (such as keywords, categories, or attributes) and user interaction data like ratings or history. It is especially useful when user data is limited or when we want to recommend items similar to what the user has already liked. It is commonly used in movie recommendations, news articles, and e-commerce platforms.

2. Write advantages of content-based filtering approach

- It does not require data from other users
- It works well for users with unique preferences
- It can recommend new or unseen items
- It provides personalized recommendations
- It can explain why a particular item is recommended

3. Write disadvantages of content-based filtering approach

- It is difficult to extract the right features from items
- It cannot recommend items outside the user's interest area
- It may lead to over-specialization (same type of recommendations)
- It requires detailed item data
- It does not consider preferences of other users

Colab link:
https://colab.research.google.com/drive/1aqPV9Fs7Ux3U1KA_Xmgu868UEixXtj4Z#scrollTo=TMMX-u8TB6Rs

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using content-based filtering approach	
Experiment No: 11	**Date:**	**Enrolment No:92301733057**

Program (Code):

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
df=pd.read_csv("/content/movies.csv")
#preprocessing step
df["genres"]=df["genres"].fillna("")
df["overview"]=df["overview"].fillna("")
df["content"]=df["genres"]+" "+df['overview']
#TF-IDF Vectorizer
tfidf=TfidfVectorizer(stop_words="english")
tfidf_matrix=tfidf.fit_transform(df["content"])
cosine_sim=cosine_similarity(tfidf_matrix,tfidf_matrix)
#droping the duplicates
indices=pd.Series(df.index,index=df["original_title"]).drop_duplicates()
def recommendation_movie(title,topn=5):
  if title not in indices:
    return "Movie is not found!"
  idx=indices[title]
  sim_scores=list(enumerate(cosine_sim[idx]))
  sim_scores=sorted(sim_scores,key=lambda x:x[1],reverse=True)
  sim_scores=sim_scores[1:topn+1]
  movie_indices=[i[0] for i in sim_scores]
  return df["original_title"].iloc[movie_indices]
recommendation_movie("Inception")
```

Results:

...

	original_title
2897	Cypher
4401	The Helix... Loaded
134	Mission: Impossible - Rogue Nation
1786	Flatliners
2389	Renaissance

dtype: object

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using content-based filtering approach	
Experiment No: 11	**Date:**	**Enrolment No:92301733057**

Observation:

In this experiment, a content-based recommendation system is developed using TF-IDF and cosine similarity. The movie genres and overview are combined to form content features. TF-IDF is used to convert the text into numerical vectors, and cosine similarity is applied to identify similar movies.

When "Inception" is provided as input, the system recommends movies such as Cypher, The Helix... Loaded, Mission: Impossible – Rogue Nation, Flatliners, and Renaissance based on content similarity.

Thus, the system effectively generates recommendations using content-based filtering.

Post Lab Exercise:

1. What changes you feel can be done, to improve the system that you have designed, as a part of in-house lab exercise.
   - Use more features like cast, director, keywords, and ratings instead of only genres and overview
   - Apply data cleaning and preprocessing more effectively (remove noise, stemming, lemmatization)
   - Use advanced models like Word2Vec or deep learning for better similarity
   - Implement a hybrid system (combine content-based + collaborative filtering)
   - Add user ratings or feedback to personalize recommendations
   - Improve performance using optimized similarity calculations
2. Recommend top-10 "series" using IMDB movie dataset (Link: https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows).

   Attach the code with output. Also, write your observation for the same.

   CODE:-

```
import pandas as pd

# Load dataset
df = pd.read_csv("/content/movies.csv")

# Sort by original_title (alphabetically) as no rating column is explicitly available
top_movies = df.sort_values(by="original_title", ascending=True)
```

Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To recommend an item to the user using content-based filtering approach	
Experiment No: 11	Date:	Enrolment No:92301733057

```
# Select top 10 movies
top_10_movies = top_movies[["original_title"]].head(10)

# Display result
top_10_movies
```

OUTPUT:-

	original_title
4257	#Horror
4142	$upercapitalist
3339	(500) Days of Summer
4585	...E tu vivrai nel terrore! L'aldilà
3556	10 Cloverfield Lane
2903	10 Days in a Madhouse
2739	10 Things I Hate About You
393	102 Dalmatians
3293	10th & Wolf
3504	11:14

Observation:

In this experiment, the IMDb dataset is utilized to extract and recommend the top 10 TV series based on their ratings. The dataset is first filtered to include only TV series and then sorted in descending order according to their IMDb scores.

The results highlight highly rated shows such as *Breaking Bad*, *Planet Earth*, and *Chernobyl* among the recommendations. This demonstrates that ranking content based on ratings is an effective approach for identifying and suggesting popular and high-quality series.